

HV Detection PCB – DC Link Side, located within HV Junction Box

TS **LV**

TS+ voltage divided across five 200k Ohm resistors. Power dissipation divided across string

Each resistor rated for 150V, so capable of up to 750V – above the maximum 600V TS Voltage and can handle transients

Transient Voltage Suppression. Clamps transient spikes in TS DC–Link Voltage, preventing damage to downstream components

12V Zener clamps gate voltage of Q3 to 12V, and prevents noise or low voltage accidentally triggering "HV Detected" signal until 12V gate voltage.

Starts conducting at 12V. Once voltage through D14 creates 1.24V drop across R60, U13 turns on like a low side switch.

3.3V Zener Diode D15 protect inputs to U13, clamping voltage to 3.3V

Source follower turns on when sufficient voltage on TS line. Provides necessary current to drive optocoupler LED, without exposing U13 to full TS voltage

Galvanic isolation between Tractive System (TS) and Low Voltage (LV) planes

4mm Creepage Distance on PCB with conformal coating

SCS:
– Red LED turns on when HV detected.
Safe state = off, so red LED uses HV_DETECTED signal (short to GND means off = safe state)

MOSFETs are used to amplify + invert signal

U13: TLP75A

U38: AZ431LBNTR–G1

U39: XP262N7002TR–G

U25: XP262N7002TR–G

U24: XP262N7002TR–G

U23: IPD80R4K5P7

U22: MMSZ4699T1G 12V zener

U21: MMSZ4684–E3–08

U20: 1.5SMC300CA

U19: 1.5SMC300CA

U18: 1.5SMC300CA

U17: 1.5SMC300CA

U16: 1.5SMC300CA

U15: 1.5SMC300CA

U14: 1.5SMC300CA

U13: TLP75A

U12: IPD80R4K5P7

U11: XP262N7002TR–G

U10: XP262N7002TR–G

U9: XP262N7002TR–G

U8: XP262N7002TR–G

U7: XP262N7002TR–G

U6: XP262N7002TR–G

U5: XP262N7002TR–G

U4: XP262N7002TR–G

U3: XP262N7002TR–G

U2: XP262N7002TR–G

U1: XP262N7002TR–G

U0: XP262N7002TR–G

U-1: XP262N7002TR–G

U-2: XP262N7002TR–G

U-3: XP262N7002TR–G

U-4: XP262N7002TR–G

U-5: XP262N7002TR–G

U-6: XP262N7002TR–G

U-7: XP262N7002TR–G

U-8: XP262N7002TR–G

U-9: XP262N7002TR–G

U-10: XP262N7002TR–G

U-11: XP262N7002TR–G

U-12: XP262N7002TR–G

U-13: XP262N7002TR–G

U-14: XP262N7002TR–G

U-15: XP262N7002TR–G

U-16: XP262N7002TR–G

U-17: XP262N7002TR–G

U-18: XP262N7002TR–G

U-19: XP262N7002TR–G

U-20: XP262N7002TR–G

U-21: XP262N7002TR–G

U-22: XP262N7002TR–G

U-23: XP262N7002TR–G

U-24: XP262N7002TR–G

U-25: XP262N7002TR–G

U-26: XP262N7002TR–G

U-27: XP262N7002TR–G

U-28: XP262N7002TR–G

U-29: XP262N7002TR–G

U-30: XP262N7002TR–G

U-31: XP262N7002TR–G

U-32: XP262N7002TR–G

U-33: XP262N7002TR–G

U-34: XP262N7002TR–G

U-35: XP262N7002TR–G

U-36: XP262N7002TR–G

U-37: XP262N7002TR–G

U-38: XP262N7002TR–G

U-39: XP262N7002TR–G

U-40: XP262N7002TR–G

U-41: XP262N7002TR–G

U-42: XP262N7002TR–G

U-43: XP262N7002TR–G

U-44: XP262N7002TR–G

U-45: XP262N7002TR–G

U-46: XP262N7002TR–G

U-47: XP262N7002TR–G

U-48: XP262N7002TR–G

U-49: XP262N7002TR–G

U-50: XP262N7002TR–G

U-51: XP262N7002TR–G

U-52: XP262N7002TR–G

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U-58: XP262N7002TR–G

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U-62: XP262N7002TR–G

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U-89: XP262N7002TR–G

U-90: XP262N7002TR–G

U-91: XP262N7002TR–G

U-92: XP262N7002TR–G

U-93: XP262N7002TR–G

U-94: XP262N7002TR–G

U-95: XP262N7002TR–G

U-96: XP262N7002TR–G

U-97: XP262N7002TR–G

U-98: XP262N7002TR–G

U-99: XP262N7002TR–G

U-100: XP262N7002TR–G

U-101: XP262N7002TR–G

U-102: XP262N7002TR–G

U-103: XP262N7002TR–G

U-104: XP262N7002TR–G

U-105: XP262N7002TR–G

TSAL Logic PCB – located within LV Electronics Box

(PCB Integrated with AMS/IMD latching logic)

Filtering of the signal to prevent noise causing false detection

Pull down to establish safe state in the event of disconnected wire

Flashing light operation controlled by Schmitt Trigger Oscillator.

When HV_DETECTED is true, the diode is off, and the output oscillates at required 2–5 Hz frequency, with 50% duty cycle.

When HV_DETECTED is false, the input to the second Schmitt inverter is pulled high, and the output is low.

Red light must oscillate between 2–5 Hz
Due to DC bias derating of 10uF capacitor, variable resistor used to manually tune frequency.

Voltage Divider to 5V for LED driver pin

TSAL LED PCB – located at top of Roll Hoop

Constant current LED driver
Switching converter to increase efficiency
Less thermal dissipation compared to linear regulator

Safe State is LED off

ON OFF

RED_LED_ENABLED_5V

Weak pull down to ensure safe state if wire disconnected

High Power, bright LEDs

IS HV Detected outside the TSAC?	RED LIGHT Behaviour
Yes	Flash at 2-5Hz
No	OFF

Vehicle: STAG 12

Drawn By: Adam Eastman

Checked By: Ray Wang, John McGarvey

CAD Part: N/A

SUFST

Sheet: TSAL_RED_LED

File: TSAL_RED_LED.kicad_sch

Title: TSAL RED LED

Size: A3

Date: 2026-03-24

Rev: v1.0.0

KiCad E.D.A. 9.0.4

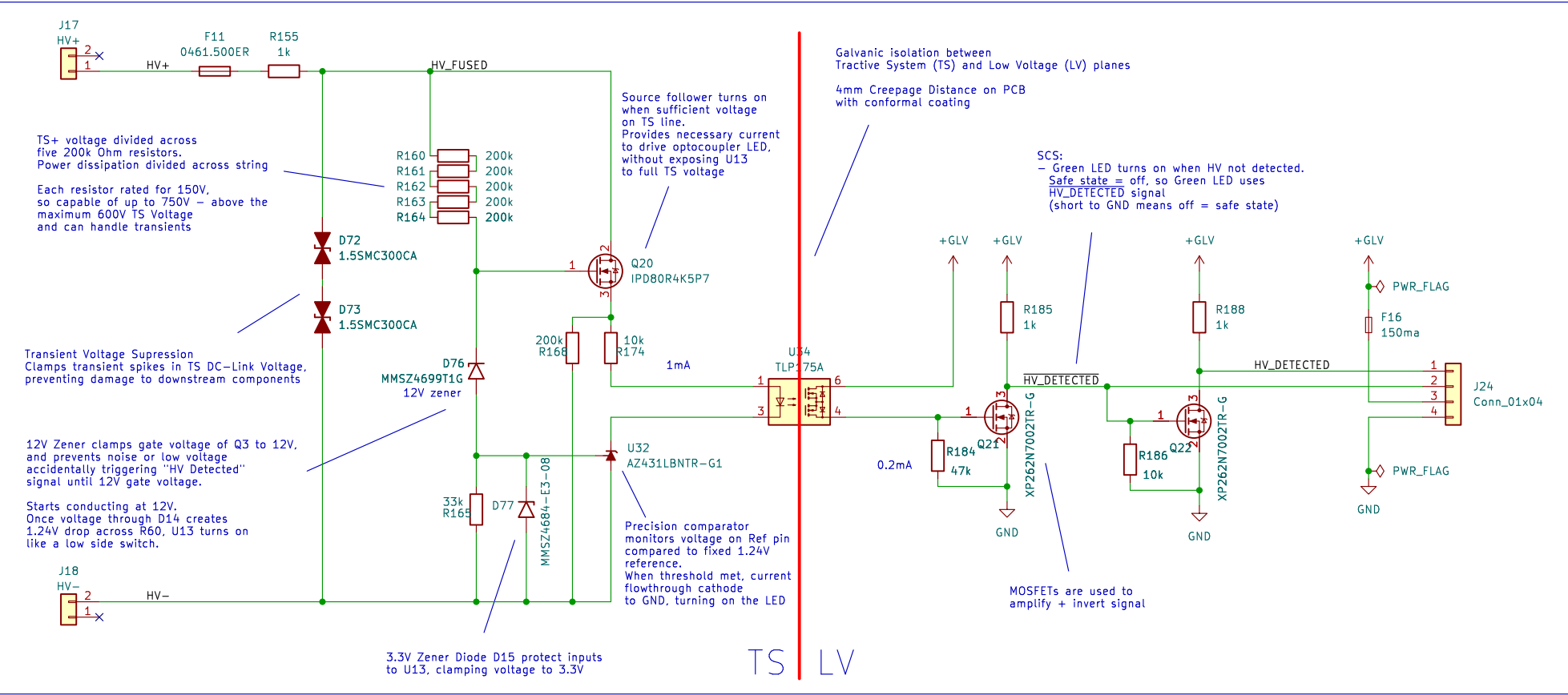
Id: 11/15

Vehicle: STAG 12
Drawn By: Adam Eastman
Checked By: Ray Wang, John McGarvey
CAD Part: N/A
SUFST
Sheet: TSAL_RED_LED
File: TSAL_RED_LED.kicad_sch

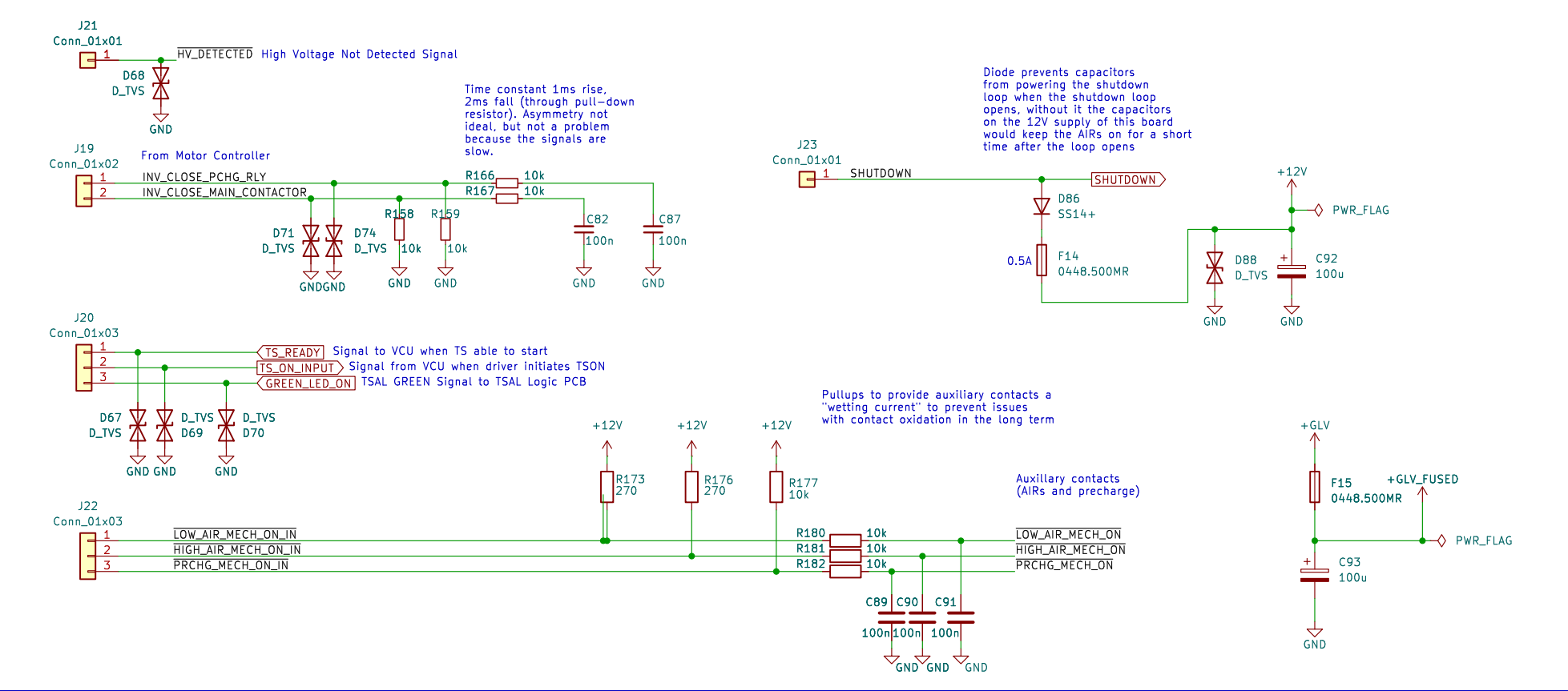
Title: TSAL RED LED

Size: A3	Date: 2026-03-24	Rev: v1.0.0
KiCad E.D.A. 9.0.4		Id: 11/15

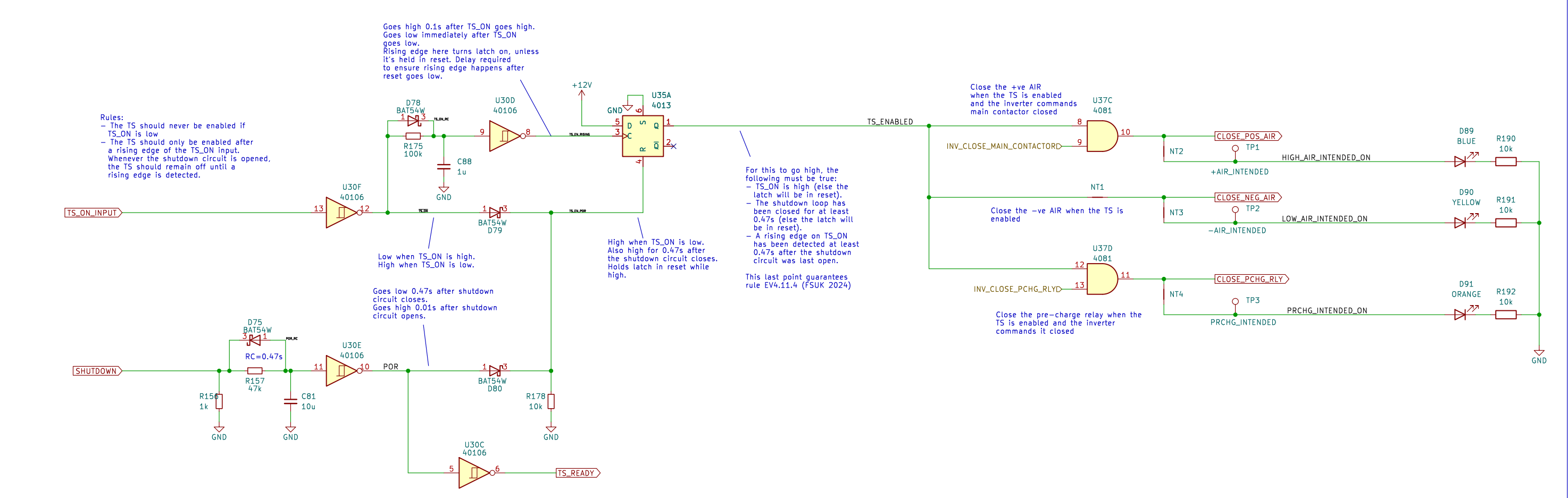
HV Detection PCB – located within Accumulator, measuring vehicle side of AIRs



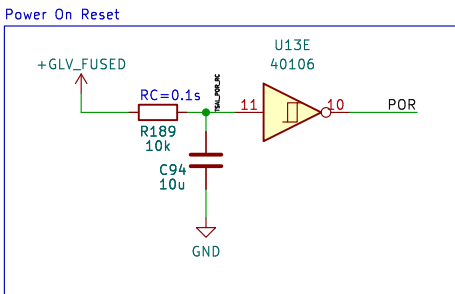
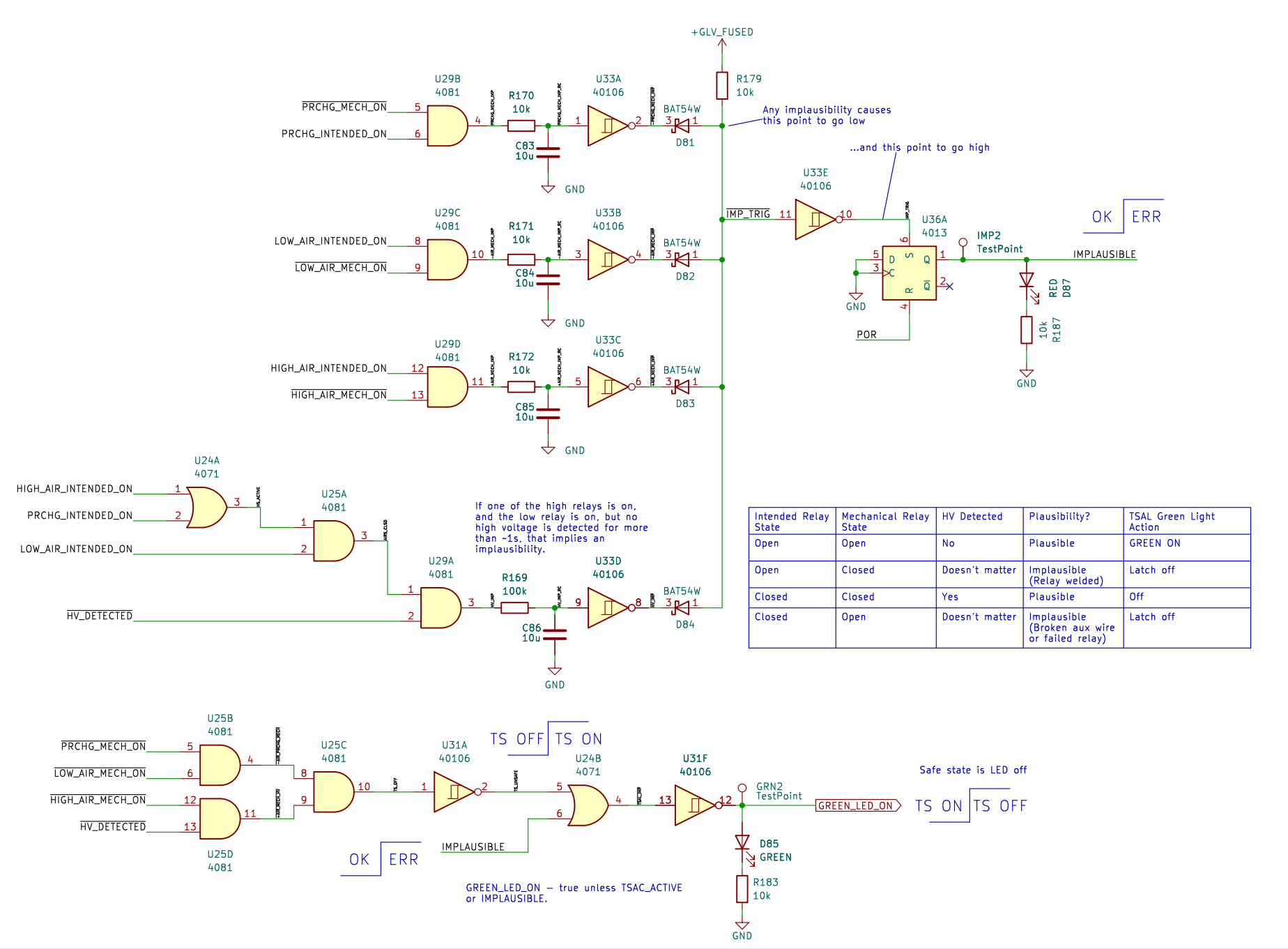
Traction Relay Controller PCB – Inputs and Outputs



Traction Relay Controller PCB – AIR Control Logic

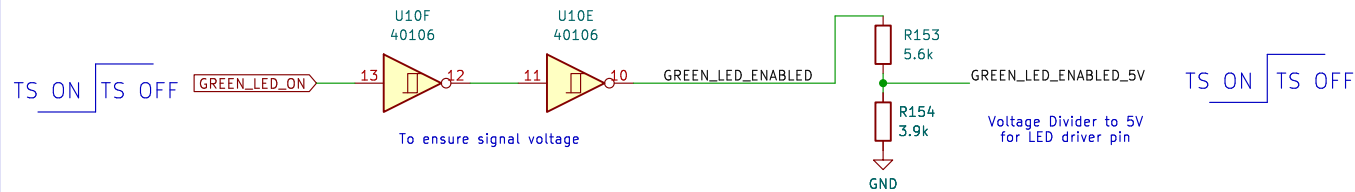


Traction Relay Controller PCB – GREEN Light Implausibility Check

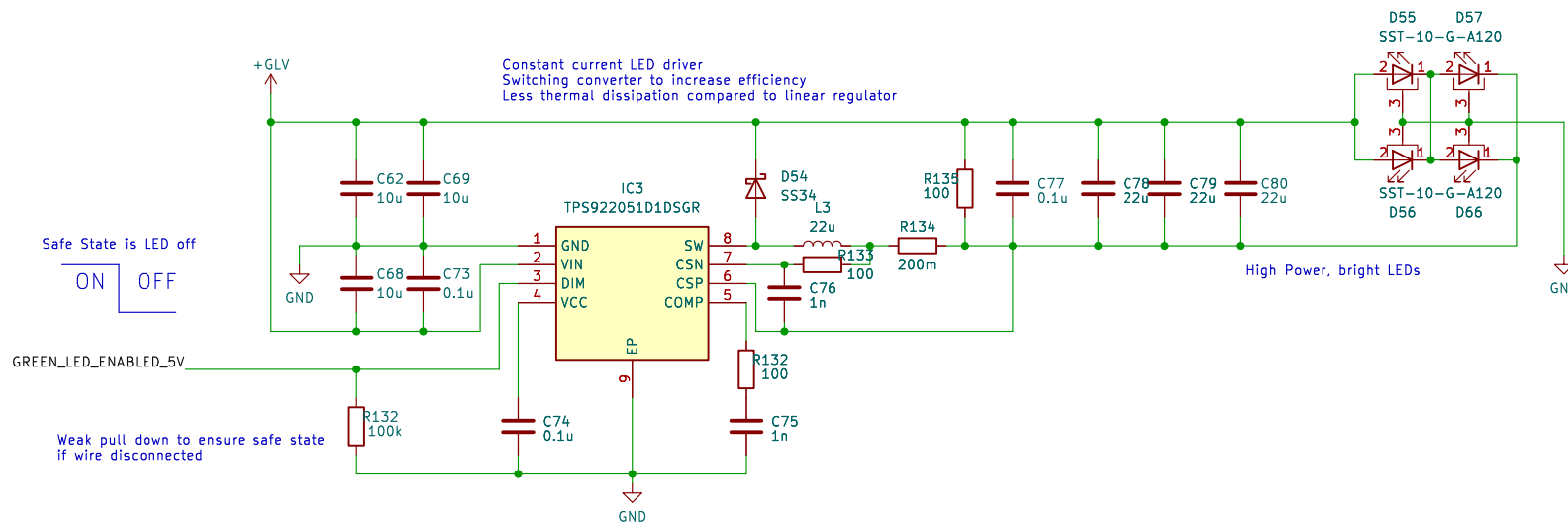


Intended Relay State	Mechanical Relay State	HV Detected	Plausibility?	TSAL Green Light Action
Open	Open	No	Plausible	GREEN ON
Open	Closed	Doesn't matter	Implausible (Relay welded)	Latch off
Closed	Closed	Yes	Plausible	Off
Closed	Open	Doesn't matter	Implausible (Broken aux wire or failed relay)	Latch off

TSAL Logic PCB – located within LV Electronics Box (PCB Integrated with AMS/IMD latching logic)



TSAL LED PCB – located at top of Roll Hoop



Vehicle: STAG 12
 Drawn By: Adam Eastman
 Checked By: Ray Wang, John McGarvey
 CAD Part: N/A

SUFST

Sheet: TSAL GREEN LED
 File: TSAL_GREEN_LED.kicad_sch

Title: TSAL GREEN LED

Size: A4 Date: 2026-03-25

KiCad E.D.A. 9.0.4

Rev: v1.0.0

Id: 15/15

